FULCRUMHQ

LEAPTHOUGHT

accenture

JTC OPTIMUS 2.0

TRAINING GUIDE – ADMINISTRATIVE ACTIONS

Module 2c

1. Configuration of PowerBI Dashboard

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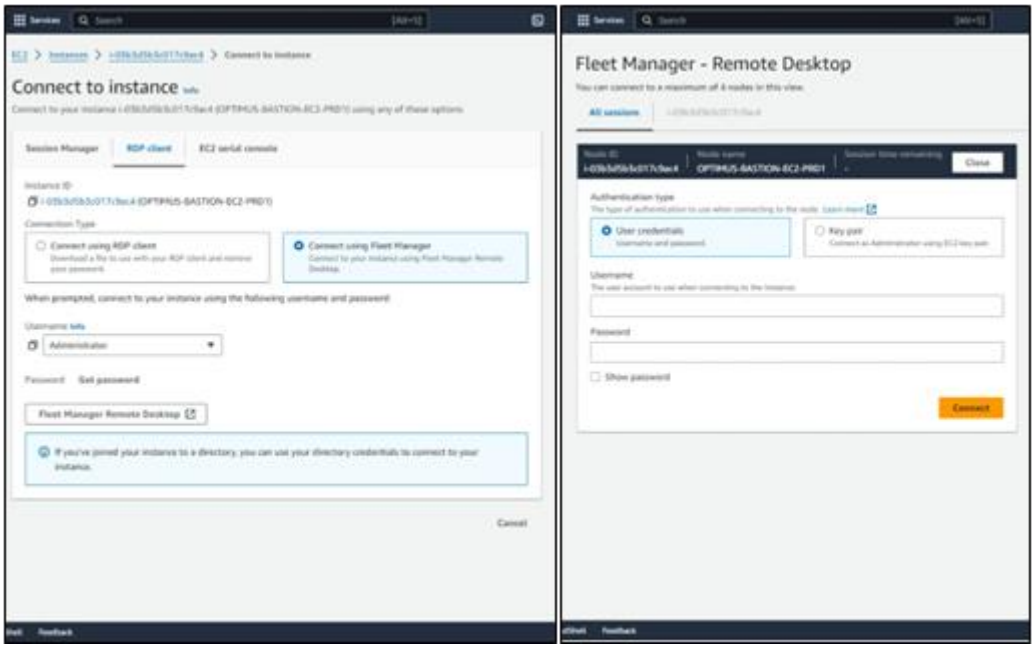
PowerBI Admin Features

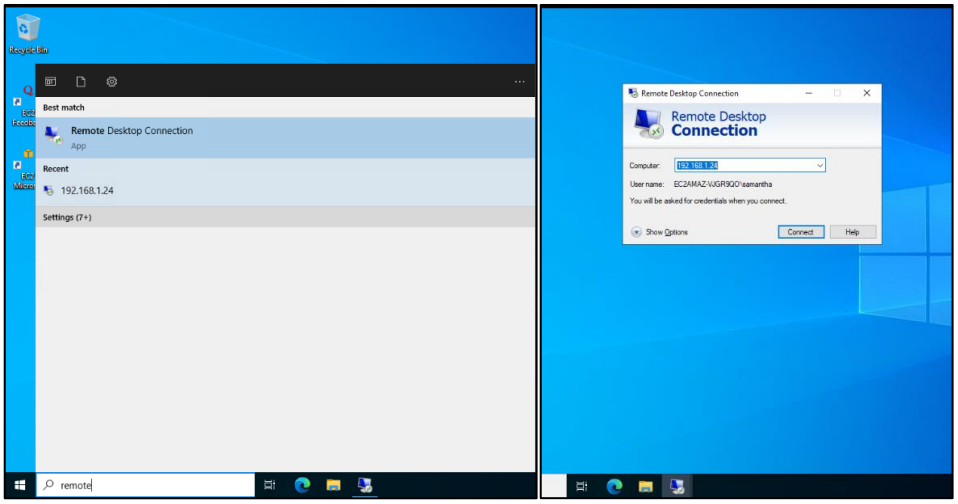
Login

1. Login to AWS Console and select **OPTIMUS-BASTION-EC2-PRD1** *(UAT Environment: OPTIMUS-BASTION-EC2-UAT1)*
2. Select **RDP Client** > **Connect Using Fleet Manager** > **Fleet Manager Remote Desktop**
3. Select and insert **User Credentials** and click **Connect**



4. Hover on top menu bar to check if IP Address for is **192.168.2.24** (*UAT Environment: 192.168.1.24)*

5. If not, open **Remote Desktop Connection**, select the above IP address and click **Connect**.



Accessing PBI Report Server

A generated token is required for each access to the Power BI Report Server.

1. Open the **TokenGenerator** application located in **C:\PBIInstaller\JTC_PBi_CustomAuth_Prod_v1.0\JTC_PBi_CustomAuth_Prod_v1.0\JTC_PBi_CustomAuth_PRD_TokenGenerator_v1.0** *(UAT Environment: C:\PBIInstaller\JTC_PBi_CustomAuth_UAT_v2.0\JTC_PBi_CustomAuth_UAT_TokenGenerator_V2.0)*
2. **Fill up the following fields**
   a. **User Id**: pbireportmgr@jtc
   b. **Expiry** (in seconds): 999999999 (maximum expiry time allowed)
3. Click **Generate Token** and copy the token generated

4. To access the Power BI Report Server, copy and replace the token url slug generated in the following link in browser: **https://optimus-bi.fulcrumhq.build/ReportServer/logon.aspx?token={token url slug**

generated} *(UAT Environment: https://optimus-qas-bi.fulcrumhq.build/ReportServer/logon.aspx?token={token url slug generated})*

Example:

Token url slug generated: AbcdEfgz123asd

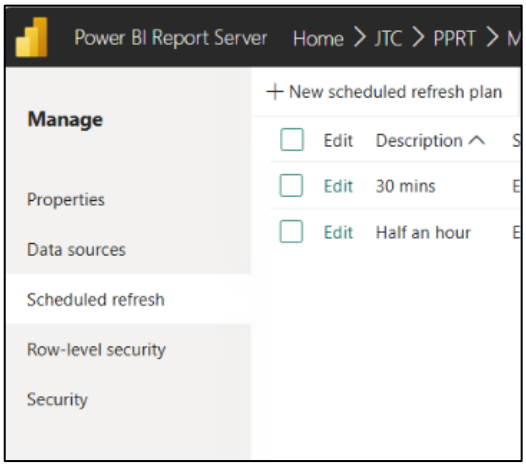
Browser URL: https://optimus-bi.fulcrumhq.build/ReportServer/logon.aspx?token= AbcdEfgz123asd

5. Repeat the above steps whenever the token expires.

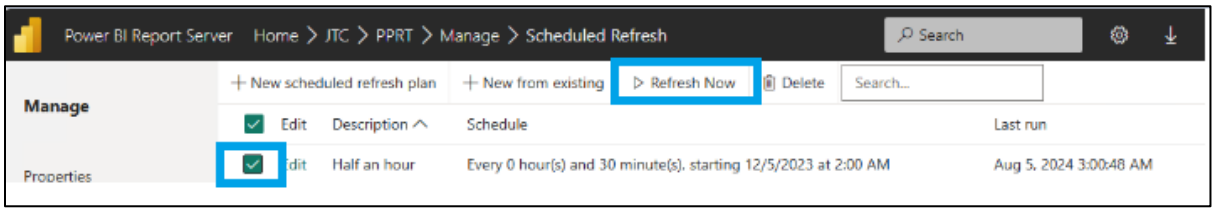
Triggering Adhoc Refreshes

1. Access the report server (refer to *Accessing PBI Report Server*)
2. Click on **JTC** folder > Select the desired dashboard and click on the ellipse dots

3. Select **Manage** and go to select **Scheduled refresh**



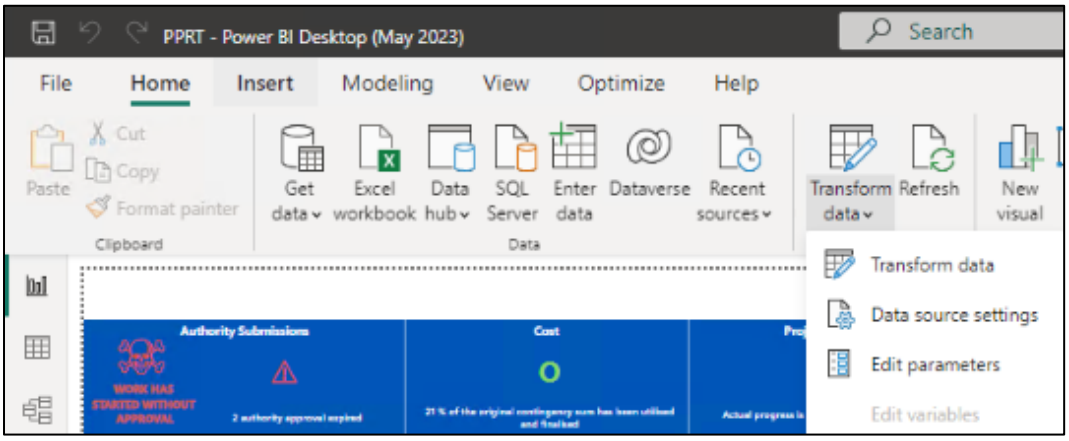
4. Click on any existing schedule > Click **Refresh Now**



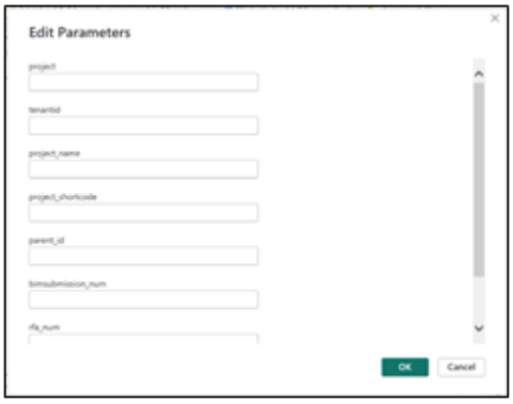
Configuring new PowerBI Report

Creating and Editing a PBI Desktop Document

1. Duplicate an existing PBIX file template
2. Edit the file as follows:
   a. Open the pbix file with PowerBI desktop *(installed in console)*
   b. **Home** > Click **Transform Data** drop down



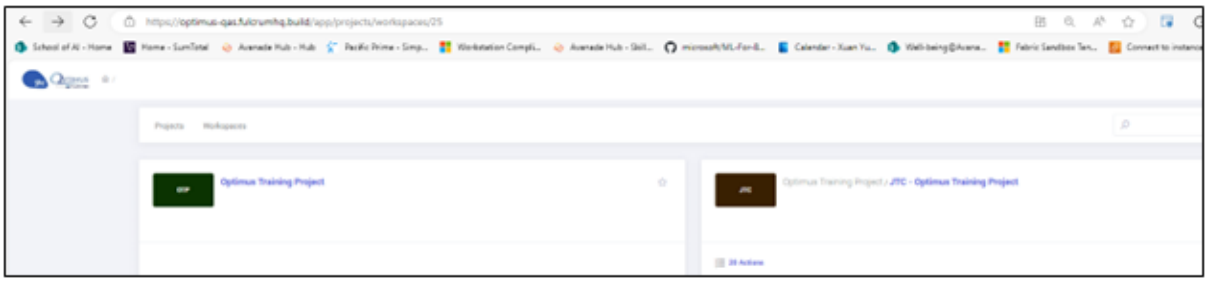
   c. Select **Edit Parameters** and input parameters accordingly (refer to *Default Parameters (for Projects Migrated from O1)* and *Variable Parameters*)



Obtaining Parameters

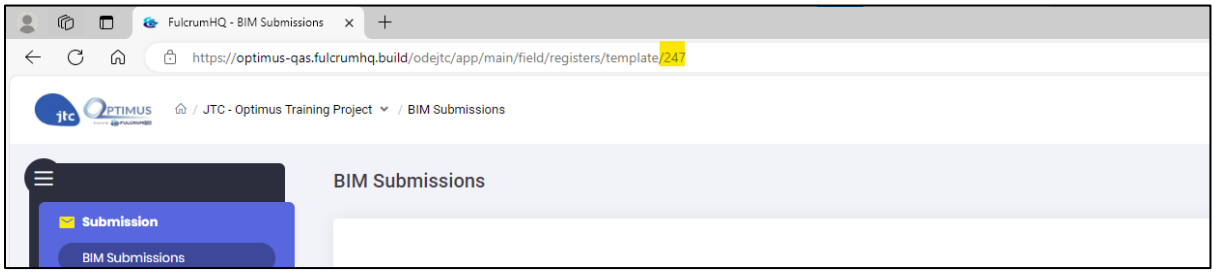
Parent ID

1. Login to OPTIMUS v2.0
2. Click into the desired project, refer to the URL and obtain the numbers after **...workspaces/**



Form ID

1. Login to OPTIMUS v2.0 > Go to desired Project
2. Click the desired form, refer to the URL and obtain the numbers after **...template/**

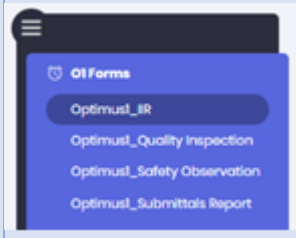


Default Parameters (for Projects Migrated from O1)

Parameters	Source	Input
so_O1_num	Optimus1_Safety Observation form in Optimus project	521
Qdi_O1_num	Optimus1_Quality Inspection form in Optimus project	522
submissions_o1_num	Optimus1_Safety Observation form in Optimus project	520
iir_O1_num	Optimus1_IIR form in Optimus project	519

project		optimus
tenantid		PROD: 1007 QAS: 1008

Variable Parameters

Parameters	Source	Comments
Project_name_o1	Optimus Project > O1 forms > Look for Project Title 	Project Title must be consistent across all 4 forms
project_name		Free Text, to align with OPTIMUS v2.0 convention
project_shortcode		O2 Project Shortcode, to align with OPTIMUS v2.0 convention
parent_id		Refer to *Obtaining Parameters*
project_details_num	Project Details form	
Contract_phases	Contract Phases form	
Soi_num	SO Instruction form	
Iir_num	Incident Investigation Report form	
So_num	Safety Observation form	
Safety_metric_num	Safety Metrics form	
Dengue_num	Dengue Reporting form	

Rac_num	Regulatory Approval Checklist form	
Qscore_num	Q-Score form	
Qdi_num	Quality/Defects Inspection form	
Bimsubmission_num	BIM Submissions form	
Rfa_num	Request for Approval (RFA) form	
Rfi_num	Request for Information (RFI) form	
Trade_demo_num	Trade Demo form	
Contractor_cashflow	Contract Cashflow form	

Publishing Dashboard in PBI Report Server

1. Access the **Power BI Report Server** *(Refer to Accessing PBI Report Server)*
2. Select the **JTC** folder
3. Click **Upload** and select the desired Power BI Desktop Document to upload

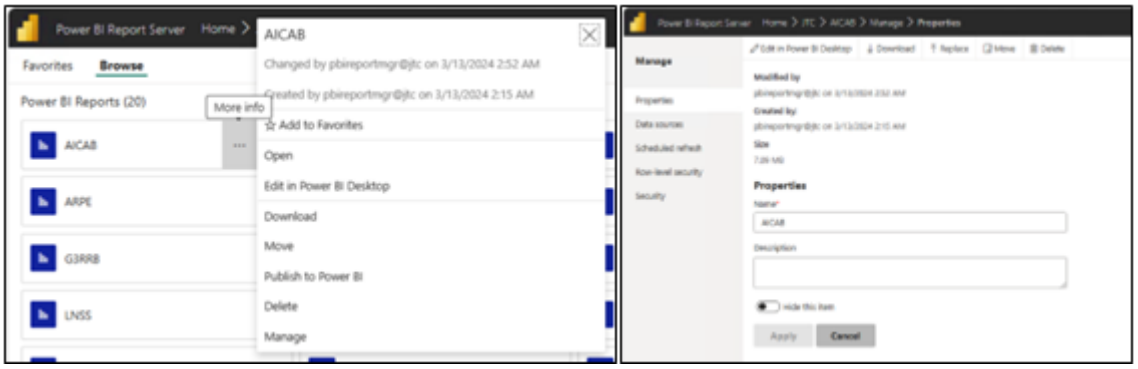
Configuring and Managing Dashboards on Report Server

All functionalities to manage the dashboard can be found in the Manage Panel.

Access Manage Panel

1. Access the **Power BI Report Server** *(Refer to Accessing PBI Report Server)*
2. Click on **JTC** folder > Select the desired dashboard and click on the ellipse dots

3. Select **Manage**

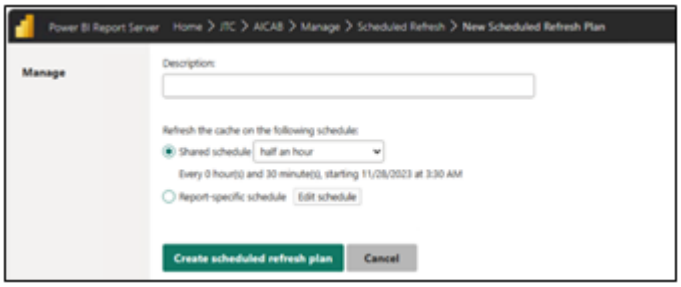


Connecting to Database

1. Under the Manage panel, select **Data sources**
2. Input the following
   a. Select **Authentication Type:** Basic Authentication
   b. Input **User name**: *To be provided separately*
   c. Input **Password**: *To be provided separately*
3. Click on **Test connection** to establish the connection. The test outcome will indicate whether it is successful.
4. If successful, Click **Save**. If not, check if the User name and Password was incorrectly typed.

Creating Scheduled Refresh

1. Under the Manage panel, select **Scheduled refresh**
2. Click on **+ New Scheduled Refresh Plan** > **Shared schedule** > **Select a predefined schedule** > **Create scheduled refresh plan**

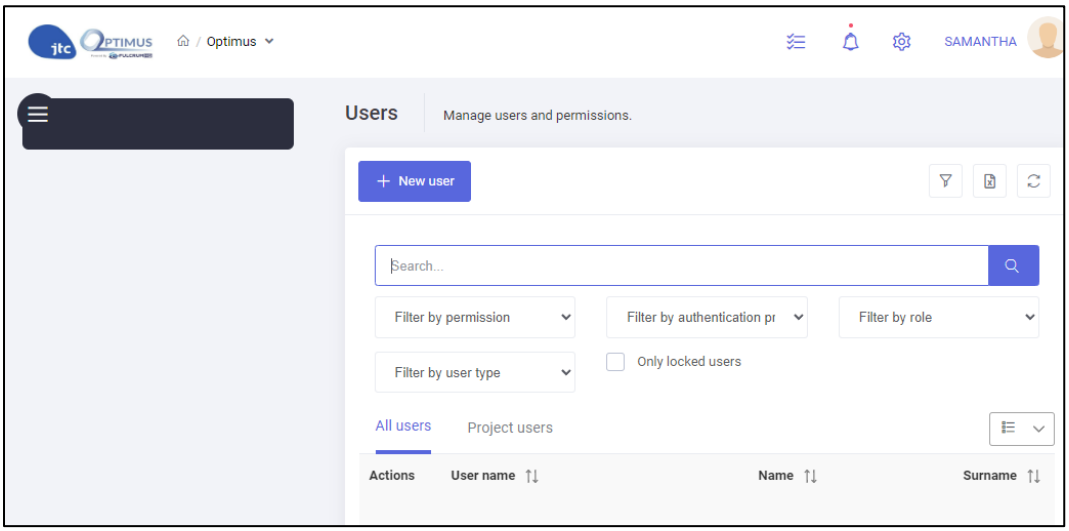


Providing User Access to PBI

Obtaining Usernames in O2

Usernames serve as the basis to identify and manage User access to Power BI.

1. In OPTIMUS v2.0, navigate to the Project Level Workspace
2. Navigate to the Users Page and obtain the User name



Adding Users to PBI Directory

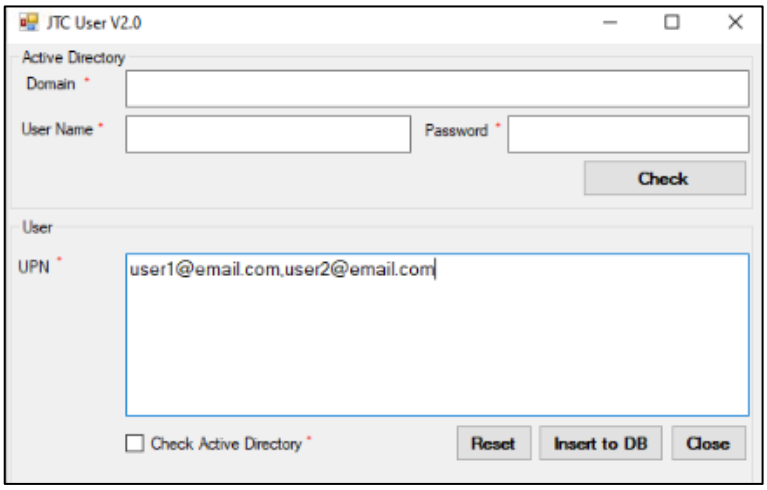
This step applies to new users that have not been added to the directory.

1. Open application in **C:\PBIInstaller\JTC_PBi_CustomAuth_Prod_v1.0\JTC_PBi_CustomAuth_Prod_v1.0\JTC_PBi_CustomAuth_PRD_UserAdminTool_v1.0\JTC User**

(UAT Environment: C:\PBIInstaller\JTC_PBi_CustomAuth_UAT_v2.0\JTC_PBi_CustomAuth_UAT_v2.0\JTC_PBi_CustomAuth_PRD_UserAdminTool_v2.0\JTC User)

2. Under **User**,
   a. **Check Active Directory**: Uncheck
   b. **UPN:** Insert O2 usernames *(refer to Obtaining Usernames in O2)*

3. To add multiple users, delineate with "," with no additional spacing i.e. to add user1@email.com and user2@email.com, input 'user1@email.com ,user2@email.com'

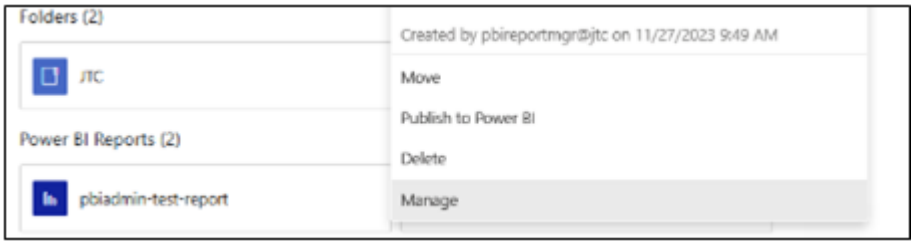


4. Click **Insert to DB**

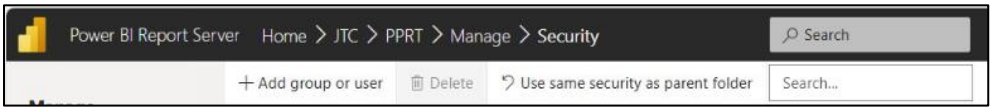
Providing User Access to Dashboard

This step allows us to provide access to users to view the Power BI Dashboards within each project, based on their O2 usernames. If there are issues with adding the users, refer to *Adding Users to PBI Directory*.

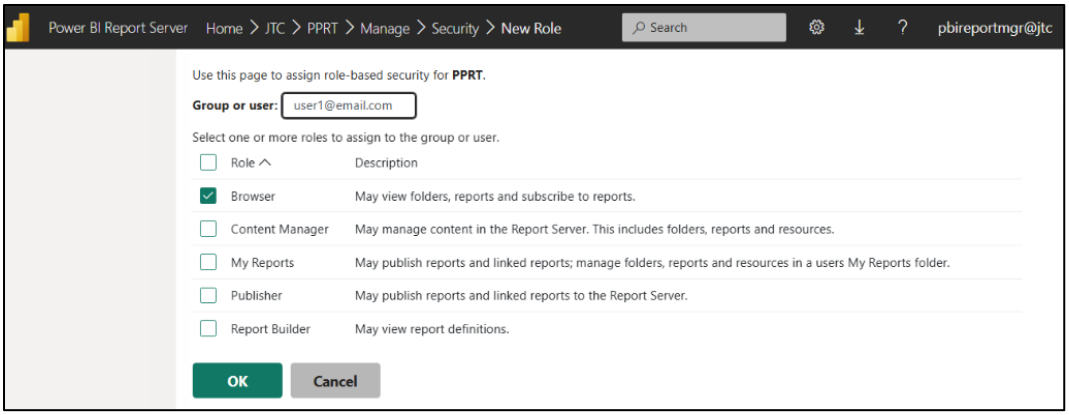
Note: Skip this step for the JTC DBE, Safety and CQD team, who are added to projects by default. New JTC DBE, Safety and CQD users, are required to be added to the list of users into the JTC folder and existing projects



1. On the Manage pane, select **Security**
2. For new projects, click **Use same security as parent folder**. It will inherit the security for default users i.e. JTC DBE, Safety and CQD team



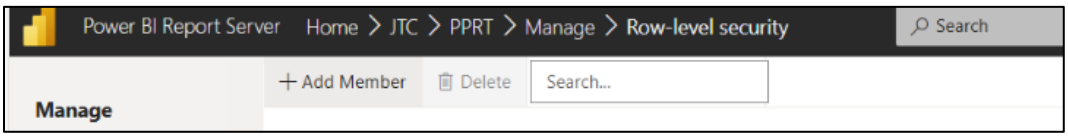
3. On the top left corner of the table, select **Add group or user**
4. Insert the **Username** individually, check the **Browser** option and click **OK**



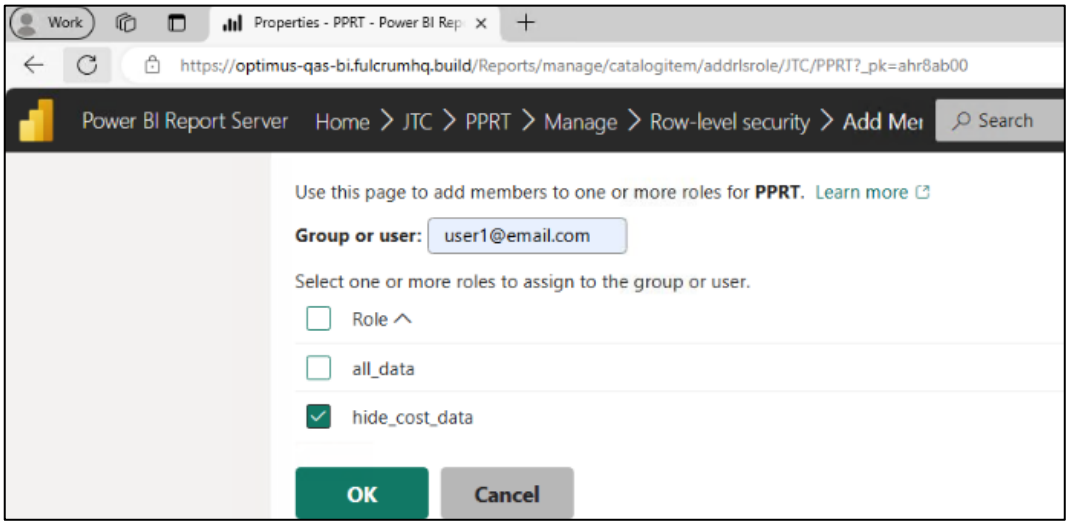
Providing User Access to Data in Dashboard

This step determines the data users can see on the dashboard.

1. On the Manage pane, select **Role-level security**
2. On the top left corner of the table, select **Add Member**

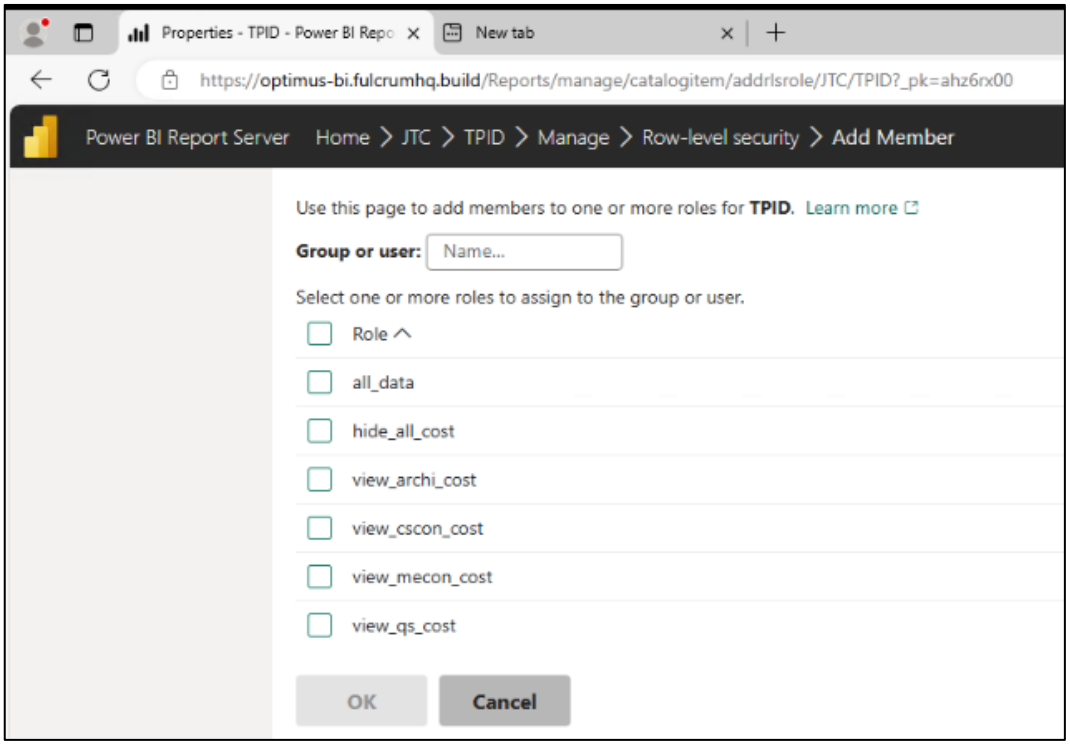


3. Insert the **Username** individually, check the roles that apply, and click **OK**
   a. **hide_cost_data**: RSS, Contractor, Quality Surveyor
   b. **all_data**: Architect, QP Architect, C&S Consultant, M&E Consultant, JTC



4. For projects that have multiple contracts, there will be following roles:
   a. **hide_cost_data**: Users who should not see any cost data
   b. **all_data**: Users who can see all contract
   c. **view_archi_cost**: Users who can view Architect contract
   d. **view_cscon_cost**: Users who can view C&S Consultant contract
   e. **view_mecon_cost**: Users who can view M&E Consultant contract

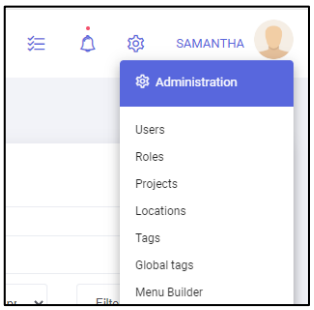
f. **view_qs_cost:** Users who can view Quality Surveyor contract



Configuring Dashboard on FulcrumHQ

Setting up connection to PBI URL in FulcrumHQ

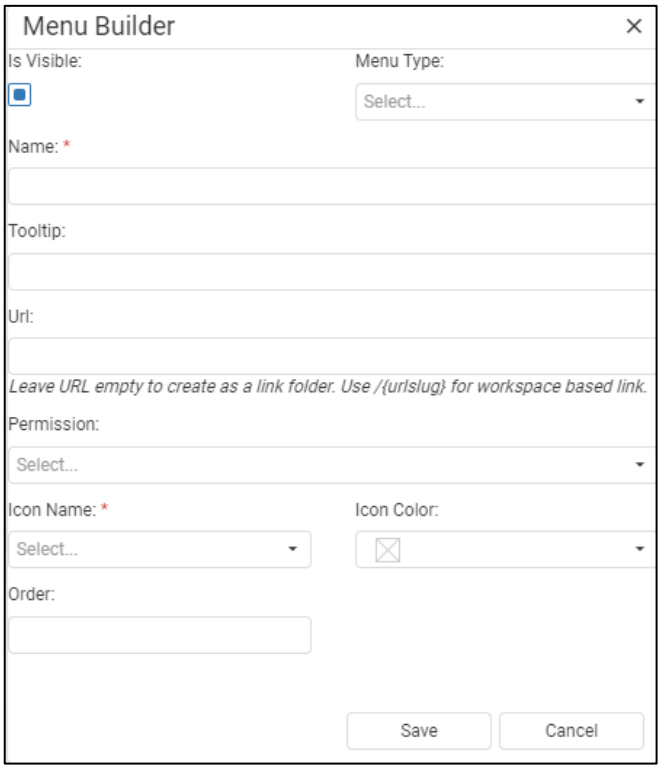
1. Access **Administration** > **Menu Builder**



2. Click the **+** icon to create a new menu item

3. Input the parameters and click **Save** *(refer to Inputs for PBI connection to FulcrumHQ)*

1. **Is Visible**: Yes
2. **Name**: Project Dashboards
3. **URL**: *(refer to Inputs for PBI connection to FulcrumHQ)*
4. **Menu Type**: Link
5. **Icon Name**: *(refer to Inputs for PBI connection to FulcrumHQ)*
6. **Icon Colour**: *(refer to Inputs for PBI connection to FulcrumHQ)*
7. **Menu Order:** A relatively smaller number means a higher position in the menu. Provide a suitable number



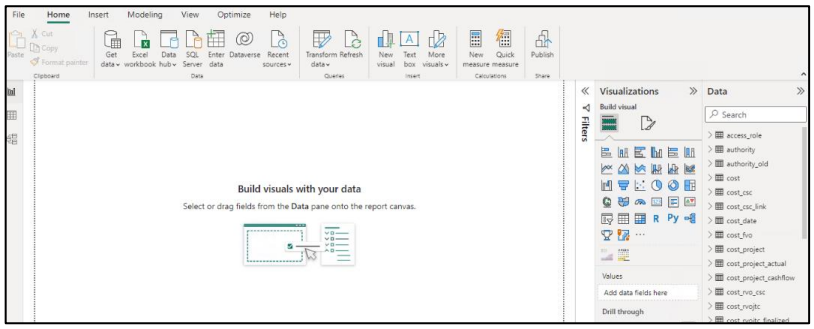
Inputs for PBI connection to FulcrumHQ

Parameters	Project Dashboards
URL	https://optimus.fulcrumhq.build/api/services/app/Reporting/PowerBIReportServer/redirect?Ou={urlslug}&Role=viewer&ReportName=powerbi/JTC/project_shortcode?rs:embed=true
Icon Name	widgets
Icon Color:	#f2dd8f

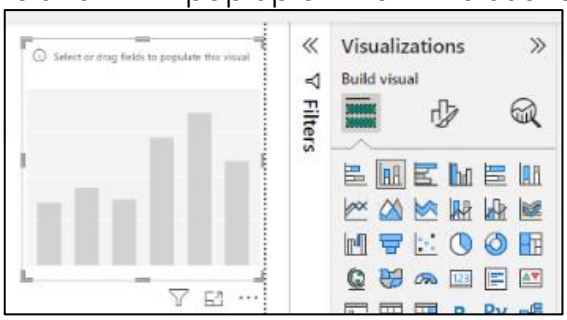
Customising Power BI Dashboards

Create Visualisation

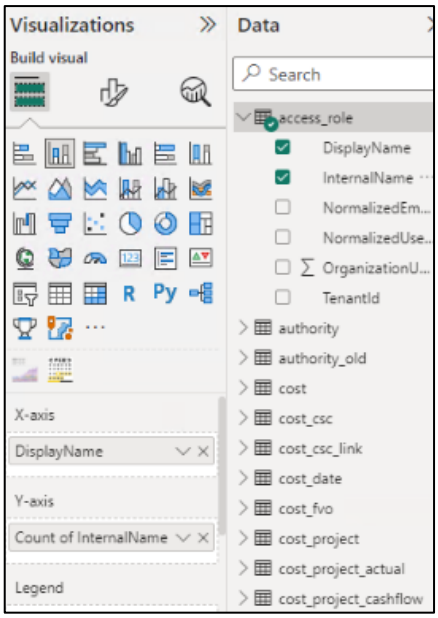
1. On the left panel, navigate to **Visualizations** pane



2. To add new visuals in the dashboard, go select any desired visualizations in **Visualizations** pane and it will pop up on the white dashboard

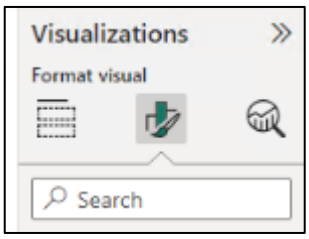


3. Pull any columns under **Data** pane to fill up the parameters in **Visualizations** pane



4. For existing visuals, you may shift and resize them via the Dashboard

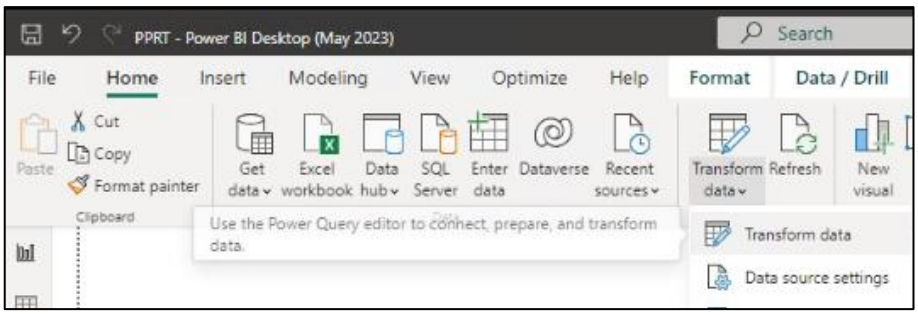
5. Use the **Visualization** panel to edit the appearance and content of chart types, legends, labels and more



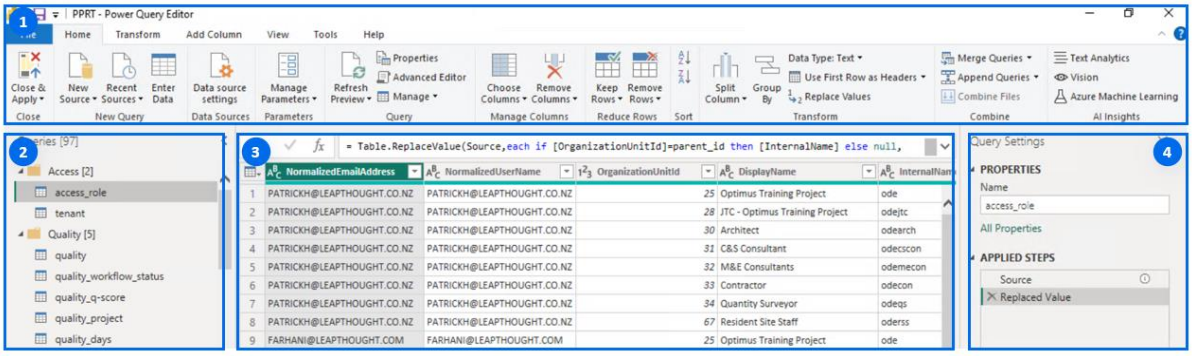
Detailed instructions: [Visualization types in Power BI - Power BI | Microsoft Learn]

Transform Data and Queries

1. **Home** > **Transform data** to launch Power Query Editor

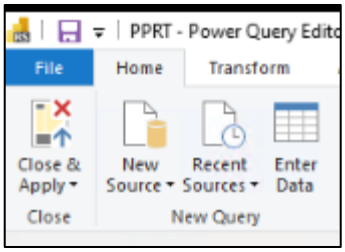


2. Standard Functionalities in Power Query Editor



1. **Home Pane**: Predefined functionalities and transformations steps that can be applied to data by clicking
2. **Queries Pane**: All tables that will be loaded and can be used for visualizations. Queries extracts the data from database powered by SQL
3. **Preview Table**: Preview the output table after going through transformation
4. **Query Settings**: Steps that are applied to manipulate the data using M Code can be found here

3. Click **Close & Apply** to save all transformations steps



Detailed instructions: [Power Query documentation - Power Query | Microsoft Learn]